

More general Gaussian likelihoods: GGaussian and GGaussianS

Parameterisation

The Gaussian distribution is

$$f(y) = \frac{\sqrt{\tau}}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\tau(y - \mu)^2\right)$$

for continuously responses y where

μ : is the the mean

τ : is the precision

These likelihood families generalise the normal procedure. We allow for two linear predictors in the model. One needs to be “simple”, only consists of fixed effects, while the other one is general and defined via the formula.

For family=“ggaussian”, then

$$\text{link}(\mu) = \text{formula} \quad \text{link.simple}(\tau) = \text{simple}$$

while for family=“ggaussianS”, then

$$\text{link}(\tau) = \text{formula} \quad \text{link.simple}(\mu) = \text{simple}$$

The default link is the *identity* for the mean and the *log* for the precision. We will describe each model separately, as the specifications are a little different.

Family *ggaussian*

Link-function

The mean is given by the linear predictor η from the formula

$$\text{link}(\mu) = \eta$$

where the default is the *identity*-link. The precision is given as

$$\text{link.simple}\left(\frac{1}{s}\tau\right) = \beta_1 z_1 + \beta_2 z_2 + \dots + \beta_m z_m, \quad m \leq 10$$

where s is a **fixed** scaling (or often log-offset) and similar to the argument *scale* for family=“gaussian”. For the log-link, the precision is expressed as

$$\tau = s \times \exp(\beta_1 z_1 + \beta_2 z_2 + \dots + \beta_m z_m)$$

Note: there is no default intercept in the simple model, so it is common to set $z_1 = 1$.

Hyperparameters

The hyperparameters in the model are $\theta_1 = \beta_1, \dots, \theta_m = \beta_m$, and the priors are defined on $\theta_1, \dots, \theta_m$.

Note: The default prior for θ_1 is similar to the default prior for the precision for the Gaussian family, please change if you do not want this feature.

Specification

- `family="ggaussian"`
- Required arguments: **inla.mdata**-object that defines the response, s and covariates. The **inla.mdata** object is defined as

$$\text{inla.mdata}(y, s, z_1, \dots, z_m)$$

where each argument are vectors of the same length.

- **Note:** The scaling argument s **MUST** be given as it has no default value. Often we can just use $s = 1$ as it will auto-expand.
- **Note:** $m = 0$ is allowed, which means that $\tau = s$ with with log link.simple.

Hyperparameter spesification and default values

```
family="ggaussian"
```

```
doc Generalized Gaussian
```

```
hyper
```

```
theta1
```

```
hyperid 66501
name beta1
short.name beta1
output.name beta1 for ggaussian observations
output.name.intern beta1 for ggaussian observations
initial 4
fixed FALSE
prior normal
param 9.33 0.61
to.theta function(x) x
from.theta function(x) x
```

```
theta2
```

```
hyperid 66502
name beta2
short.name beta2
output.name beta2 for ggaussian observations
output.name.intern beta2 for ggaussian observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x
```

```
theta3
```

```
hyperid 66503
```

```

name beta3
short.name beta3
output.name beta3 for ggaussian observations
output.name.intern beta3 for ggaussian observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x
theta4
  hyperid 66504
  name beta4
  short.name beta4
  output.name beta4 for ggaussian observations
  output.name.intern beta4 for ggaussian observations
  initial 0
  fixed FALSE
  prior normal
  param 0 10
  to.theta function(x) x
  from.theta function(x) x
theta5
  hyperid 66505
  name beta5
  short.name beta5
  output.name beta5 for ggaussian observations
  output.name.intern beta5 for ggaussian observations
  initial 0
  fixed FALSE
  prior normal
  param 0 10
  to.theta function(x) x
  from.theta function(x) x
theta6
  hyperid 66506
  name beta6
  short.name beta6
  output.name beta6 for ggaussian observations
  output.name.intern beta6 for ggaussian observations
  initial 0
  fixed FALSE
  prior normal
  param 0 10

```

```

    to.theta function(x) x
    from.theta function(x) x
theta7
    hyperid 66507
    name beta7
    short.name beta7
    output.name beta7 for ggaussian observations
    output.name.intern beta7 for ggaussian observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta8
    hyperid 66508
    name beta8
    short.name beta8
    output.name beta8 for ggaussian observations
    output.name.intern beta8 for ggaussian observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta9
    hyperid 66509
    name beta9
    short.name beta9
    output.name beta9 for ggaussian observations
    output.name.intern beta9 for ggaussian observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta10
    hyperid 66510
    name beta10
    short.name beta10
    output.name beta10 for ggaussian observations
    output.name.intern beta10 for ggaussian observations

```

```

initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

```

```
survival FALSE
```

```
discrete FALSE
```

```
link default identity
```

```
link.simple default log
```

```
pdf ggaussian
```

Example

```

n <- 1000
x <- rnorm(n)
xx <- rnorm(n)
eta <- 0.1 + 1.1 * x + 2.2 * xx

s <- runif(n)
z <- rnorm(n)
zz <- rnorm(n)
eta.prec <- 1 + 0.55 * z + 1.1 * zz

y <- eta + 1/sqrt(s * exp(eta.prec)) * rnorm(n)
Y <- inla.mdata(y, s, 1, z, zz)
r <- inla(Y ~ 1 + x + xx,
          data = list(Y = Y, x = x, xx = xx, z = z, zz = zz, s = s),
          family = "ggaussian")
summary(r)

```

Family *ggaussianS*

Link-function

This is the swapped version. The mean is given by a simple linear predictor

$$\text{link.simple}(\mu) = \text{off} + \beta_1 z_1 + \beta_2 z_2 + \dots + \beta_m z_m, \quad m \leq 10$$

where “off” is a fixed offset and where `link.simple` is the *identity*-link by default. **Note:** there is no default intercept in the simple model, so it is common to set $z_1 = 1$. The precision is specified in the formula

$$\text{link}(\tau) = \text{formula}$$

using the log-link as default.

Hyperparameters

The hyperparameters in the model are $\theta_1 = \beta_1, \dots, \theta_m = \beta_m$, and the priors are defined on $\theta_1, \dots, \theta_m$.

Specification

- `family="ggaussianS"`
- Required arguments: **inla.mdata**-object that defines the response, the offset “off” and covariates. The **inla.mdata** object is defined as

$$\text{inla.mdata}(y, \text{off}, z_1, \dots, z_m)$$

where each argument are vectors of the same length.

- **Note:** The offset argument **MUST** be given as it has no default value. Often we can just use `off= 0` as it will auto-expand.
- **Note:** $m = 0$ is allowed, which means that $\mu = \text{off}$ with with identity `link.simple`.

Hyperparameter spesification and default values

`family="ggaussianS"`

doc Generalized GaussianS

hyper

theta1

hyperid 66601

name beta1

short.name beta1

output.name beta1 for ggaussianS observations

output.name.intern beta1 for ggaussianS observations

initial 0

fixed FALSE

prior normal

param 0 0.001

to.theta function(x) x

from.theta function(x) x

theta2

hyperid 66602
name beta2
short.name beta2
output.name beta2 for ggaussianS observations
output.name.intern beta2 for ggaussianS observations
initial 0
fixed FALSE
prior normal
param 0 0.001
to.theta function(x) x
from.theta function(x) x

theta3

hyperid 66603
name beta3
short.name beta3
output.name beta3 for ggaussianS observations
output.name.intern beta3 for ggaussianS observations
initial 0
fixed FALSE
prior normal
param 0 0.001
to.theta function(x) x
from.theta function(x) x

theta4

hyperid 66604
name beta4
short.name beta4
output.name beta4 for ggaussianS observations
output.name.intern beta4 for ggaussianS observations
initial 0
fixed FALSE
prior normal
param 0 0.001
to.theta function(x) x
from.theta function(x) x

theta5

hyperid 66605
name beta5
short.name beta5
output.name beta5 for ggaussianS observations
output.name.intern beta5 for ggaussianS observations
initial 0
fixed FALSE

```

    prior normal
    param 0 0.001
    to.theta function(x) x
    from.theta function(x) x
theta6
    hyperid 66606
    name beta6
    short.name beta6
    output.name beta6 for ggaussianS observations
    output.name.intern beta6 for ggaussianS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 0.001
    to.theta function(x) x
    from.theta function(x) x
theta7
    hyperid 66607
    name beta7
    short.name beta7
    output.name beta7 for ggaussianS observations
    output.name.intern beta7 for ggaussianS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 0.001
    to.theta function(x) x
    from.theta function(x) x
theta8
    hyperid 66608
    name beta8
    short.name beta8
    output.name beta8 for ggaussianS observations
    output.name.intern beta8 for ggaussianS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 0.001
    to.theta function(x) x
    from.theta function(x) x
theta9
    hyperid 66609
    name beta9
    short.name beta9

```

```

    output.name beta9 for ggaussianS observations
    output.name.intern beta9 for ggaussianS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 0.001
    to.theta function(x) x
    from.theta function(x) x
theta10
    hyperid 66610
    name beta10
    short.name beta10
    output.name beta10 for ggaussianS observations
    output.name.intern beta10 for ggaussianS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 0.001
    to.theta function(x) x
    from.theta function(x) x

```

survival FALSE

discrete FALSE

link default log

link.simple default identity

pdf ggaussian

Example

```

n <- 1000
x <- rnorm(n)
xx <- rnorm(n)
off <- runif(n)
z <- rnorm(n)
zz <- rnorm(n)

mean <- off + 0.1 + 1.1 * z + 2.2 * zz
prec <- exp(1 + 0.55 * x + 1.1 * xx)

y <- mean + (1/sqrt(prec)) * rnorm(n)
Y <- inla.mdata(y, off, 1, z, zz)
r <- inla(Y ~ 1 + x + xx,
          data = list(Y = Y, off = off, x = x, xx = xx, z = z, zz = zz),
          family = "ggaussianS")
summary(r)

```